My Math Pocket Card

Y2 · 3rd term · test on 18 November

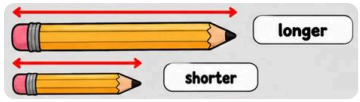
1 Measures · 2 Perimeter · 3 3D solids · 4 Data

FRONT

1 Words to compare and measure

Length is how long something is, from one end to the other. A pencil has a length.

long · short

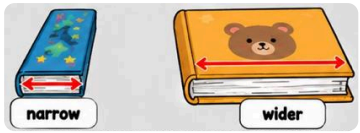


Long means how much something measures from one end to the other.

Short means not as long.

The top pencil is long. The bottom pencil is short.

wide · narrow

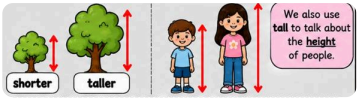


Wide means how much something measures across.

Narrow means not wide.

The big book is wide.

tall · height



Tall means how high something is, from the bottom to the top.

Height is how tall someone or something is. We also use tall for people.

The big tree is tall. The girl is tall.

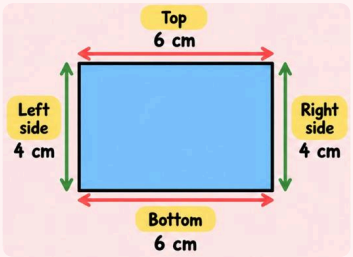
Compare two things: longer · shorter · wider · taller. ***The top pencil is longer than the bottom pencil.***

We measure length in **centimetres (cm)**, **metres (m)** and **kilometres (km)**.

2 Perimeter

distance around

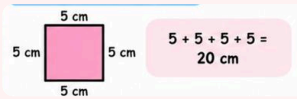
Perimeter is the total distance around a shape. **Around** means along all the sides of the shape. A **side** is one of the straight lines that make the outline of a flat shape.



Add all the sides!

Top	6 cm
Bottom	6 cm
Left side	4 cm
Right side	4 cm

6 + 6 + 4 + 4 = 20 cm

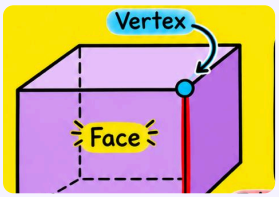


- Always write the number **and** the unit: **20 cm**.
- **Show your work**: write the sum, like 6 + 6 + 6 = 18 cm.

3 3D solids

faces · edges · vertices

A **3D solid** is a solid shape that has faces, edges and vertices.



face: a flat (straight) surface of a 3D solid. A cube has 6 faces.

edge: a straight line where two faces meet. A cube has 12 edges.

vertex: a corner where two or more edges meet. More than one vertex: **vertices**. A cube has 8 vertices.

Some solids, like the sphere, the cylinder and the cone, have a **curved surface**: a curved surface is round, not flat.



cube
6 faces
12 edges
8 vertices



cuboid
6 faces · 12 edges
8 vertices
like a cube, but the edges are not all the same length



pyramid
5 faces
8 edges
5 vertices



cylinder
2 flat faces
1 curved surface



cone
1 flat face
1 curved surface



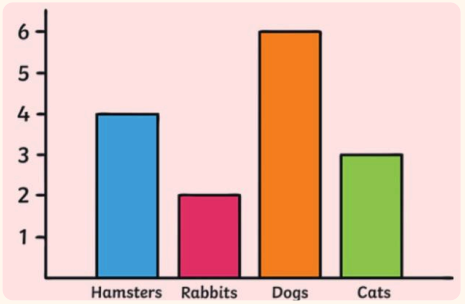
sphere
1 curved surface
no flat face, no edges, no vertices

Roll means turn over and over and move away, like a ball. A solid with a curved surface can roll: the sphere rolls. A cube has only flat faces: it cannot roll.

4 Data and bar graphs

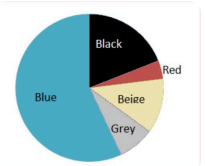
Data is all the information we collect and find out. We organize it using charts and graphs. A **piece of data** is one simple fact: ***My favorite color is blue.*** ***I have 2 pets.***

A **bar graph** shows data with bars. The taller the bar, the more there are. Look at the top of the bar and read the number on the side.



Hamsters	Rabbits	Dogs	Cats
4	2	6	3

- The most: **Dogs (6)**.
- The least: **Rabbits (2)**.
- How many cats? **3**.



The round picture is a **chart** too: it also shows data.

Make a bar graph: look at the table with numbers and colour each bar up to the right number.

5 Probability

likely · unlikely

Probability is thinking about if something will happen, won't happen, or will maybe happen.

certain	We know it will happen. ***I will eat dinner tonight.***
likely	Big chance to happen. ***You will find students in the school.***
possible	It will maybe happen. ***I will eat spaghetti for dinner.***
unlikely	Small chance to happen. ***You will see a penguin in the desert.***
impossible	We know it won't happen. ***I will eat rocks for dinner.***

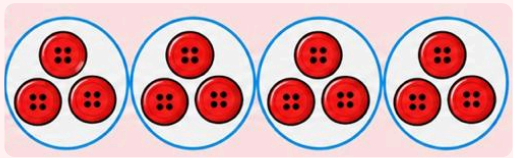
Say why with **because**: "It is impossible because ..."

6 Multiplication

making equal groups

Multiplication is a shorter way to show repeated addition. We use it to find the total when we have **equal groups**.

equal groups	have the same amount in each group.
repeated addition	is adding the same number again and again: 5 + 5 + 5.
multiplication sentence	is the math written with numbers and signs: 4 × 3 = 12.



3 + 3 + 3 + 3 = 12 → **4 × 3 = 12**

4 how many groups

3 how many in each group

4 groups of 3 buttons are 12 buttons altogether (in total).



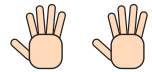
5 + 5 + 5 = 15 → **3 × 5 = 15**

3 groups of 5 flowers



2 + 2 + 2 + 2 + 2 = 10 → **5 × 2 = 10**

5 groups of 2 cookies



5 + 5 = 10 → **2 × 5 = 10**

2 hands, 5 fingers on each hand

7 Division

equal sharing

Division is a way to make equal groups of a number. We share a total into groups that have the same amount. **Share equally** means everybody gets the same amount. A **division sentence**: 10 ÷ 2 = 5.



10 ÷ 2 = 5

Mike and Laura share 10 oranges equally: 10 oranges shared into 2 equal groups is 5 oranges in each group.



8 ÷ 2 = 4

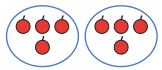
8 apples shared into 2 equal groups is 4 apples in each group.



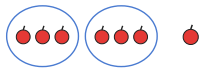
12 ÷ 3 = 4

12 cookies shared into 3 equal groups is 4 cookies in each group.

Even numbers (0, 2, 4, 6, 8, 10, 12...) can be shared equally into **2 groups**. **Odd numbers** (1, 3, 5, 7, 9...): when we share them between 2, **1 is left**.



8 between 2: 4 each



7 between 2: 3 each, 1 is left

How to write a good answer

key numbers

1. Read the question two times.
2. Start with words from the question: "*The cube has 6 faces.*"
3. Write the number **and** the unit: *18 cm*.
4. Show your work: write the math.
5. Say why with **because**.
6. Capital letter at the start, full stop at the end.

cube
6 · 12 · 8
faces · edges · vertices

pyramid
5 · 8 · 5
faces · edges · vertices

perimeter
add all the sides

certain › likely › possible › unlikely › impossible

4 × 3 = 12
12 ÷ 3 = 4

even: shared equally into 2 groups · odd: 1 is left